

NEURONEUS

The Company Brain

*On institutional memory, the scarcity of context,
and the substrate beneath the model*

A manifesto

Munich

Most organisations do not have a knowledge problem. They have a retrieval problem, a durability problem, and a trust problem, and they have been calling all three by the first name for forty years.

1 THE FORGETTING

An organisation is a machine for coordinating people. Its most valuable asset is what it collectively knows: why the architecture is shaped the way it is, which customer cannot be called before eleven, what was tried in 2023 and why it failed, who actually decides. Almost none of this is written down. It is stored in the least durable medium the organisation has available, which is people.

People leave.

Take a simple model. Let r be annual attrition, and suppose the knowledge held exclusively by one individual departs with them. The share of undocumented institutional knowledge surviving after t years is

$$S(t) = (1 - r)^t = e^{-\lambda t}, \quad \lambda = -\ln(1 - r),$$

with a half-life of

$$t_{1/2} = \frac{\ln 2}{\lambda}.$$

At a benign $r = 0.15$, the half-life of undocumented knowledge is about **4.3 years**. Not the half-life of the *company* — the half-life of the reason it does things the way it does.

This is a toy model, and it should be read as one: it assumes exclusivity of knowledge, independence of departures, and no re-derivation by those who stay. Every one of those assumptions is charitable in a direction that makes the real number worse rather than better, because the knowledge most likely to be held by exactly one person is precisely the knowledge nobody thought to write down. The model's purpose is not to produce a figure. It is to show that the decay is *exponential* and that the organisation experiences it as a series of unrelated small surprises.

Nobody notices a half-life. What they notice is that the same question gets asked again, that the answer is reconstructed from scratch, and that the reconstruction is worse than the original — and none of it is attributed to a departure two years earlier.

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2 THE COORDINATION TAX

The second cost is paid continuously rather than at exit.

A team of n people has

$$\binom{n}{2} = \frac{n(n-1)}{2}$$

pairwise channels through which state must be kept consistent. Output, at its most generous, scales linearly in n . The coordination surface scales quadratically.

Ten people maintain 45 channels. Fifty people maintain 1,225. Five times the headcount, twenty-seven times the surface. This is Brooks's observation, and it has not stopped being true because the tooling got prettier.

The status meeting is the interest payment on a debt of un-externalised state.

Every recurring alignment ritual exists because the system of record cannot answer a question that someone needs answered, so a human is deployed to be the system of record for thirty minutes. Each ritual is individually defensible. Collectively they are the organisation paying, in its scarcest currency, for the absence of an artefact.

The instinct is to fight this with discipline — write more documents, update the wiki, keep the board current. The instinct is right and it does not work, for a reason worth naming precisely: *the cost of recording is borne by the person who already knows, and the benefit accrues to someone who does not exist yet.* No amount of exhortation fixes an incentive with that shape. The recording has to be a by-product of doing the work, not a tax on top of it.

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3 WHAT CHANGED

For the whole history of software, the binding constraint on automating this was the machine. It could not read a document, summarise a thread, infer that two differently-worded tickets described one problem, or notice that a roadmap had acquired a risk. Every system that tried was a system of forms, and the forms were the tax.

That constraint is gone. It went quietly, and most organisations have not restructured around its absence.

The bottleneck moved. It is no longer intelligence. It is *context*.

A frontier model with no access to your company's state is a brilliant new hire on their first morning — every morning, for ever.

This reframing is the entire thesis, so it is worth stating without metaphor. The quality of what such a system produces is not a function of model capability alone. Write it as a product rather than a sum:

$$\text{Output} \sim C \cdot Q,$$

where C is model capability and Q is the quality of the grounded, current, permissioned context it is given. The two terms multiply. C is rising steeply, is available to everyone on identical terms, and is not a differentiator for any company that is not building models. Q is near zero almost everywhere, is specific to each organisation, and is owned by nobody.

Capability has become a commodity input. Context has not. Every marginal unit of effort spent raising C is effort spent on a term someone else is already driving to the frontier for you; effort spent raising Q compounds inside your own walls and cannot be bought.

The company that recognises this first does not *adopt* AI. It restructures so that its own operation is legible to a model by construction — so that Q is high as a property of how the work is done, not as a project someone is assigned. That is what an AI-first company actually is. Everything else is a chatbot in front of a wiki that was already out of date.

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4 THE THESIS

We are building the company brain: durable institutional memory, held as structured state, retrievable by both the people who work there and the models that work for them.

Three properties, none optional.

1. **Durable.** It outlives the people. What someone knows enters the system as a by-product of doing the work — closing the task, making the decision, writing the document — rather than as a separate act of documentation that competes with the work for the same hour.
2. **Structured.** Not a search index over prose. Projects, tasks, decisions, documents and people as entities with relationships between them, so that *who is blocked on what* is a traversal rather than an inference. Search returns documents; a brain returns answers, and the difference is whether the structure was captured or has to be recovered.
3. **Bounded.** Every answer is scoped to exactly one organisation, and that scope is enforced by the substrate rather than intended by the application. Section 5 is entirely about why this is the third property and not a footnote.

The asset being built is the third thing. A company that has run on this for two years owns a clean, connected, machine-readable account of how it operates. That artefact is worth more than any feature built on top of it, and it is the only part a competitor cannot acquire by making the same API call you did.

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5 THE HARD PART IS NOT THE MODEL

Anyone can call an inference endpoint. The difficulty — and therefore the whole of the engineering — is beneath it.

5.1 *Isolation is a property, not an intention*

A system that answers questions from a company's private corpus is one forgotten where clause away from answering them from someone else's.

The rule every team writes down is *always scope the query to the tenant*. It fails for a structural reason: the omission is invisible in review. The unscoped query compiles, runs, and returns rows that look entirely plausible. There is no error to notice. The only signal is the absence of a clause that was never there.

So we do not rely on remembering. The tenant identifier is a branded type that only the session can mint, which turns a missing scope into a compile error. The database enforces row-level security that the application cannot bypass. And a canary suite asserts, per table and per endpoint, that one tenant cannot observe another — including that an unauthorised read returns *not found* rather than *forbidden*, because a status code is itself an information channel.

Three layers, none trusting the one beneath it. Design is not evidence; the canaries are what turn an intention into a property.

5.2 Retrieval has a correctness failure that presents as a taste failure

This one deserves its space, because it is the clearest example of what goes wrong when a statistical system is treated as an exact one.

Retrieval over a private corpus means an approximate nearest-neighbour search filtered to one tenant. The natural implementation searches an HNSW index and filters the results. But an HNSW search is a *greedy traversal of a navigable graph*, not an exhaustive scan: it maintains a candidate list of size ef and returns what that traversal found. The tenant filter is applied afterwards.

Let p be a tenant’s share of all indexed chunks. The expected number of candidates surviving the filter is roughly

$$\mathbb{E}[\text{survivors}] \approx ef \cdot p.$$

With $ef = 100$ and $p = 0.01$, a query asking for ten results can expect to receive *one* — while the tenant genuinely holds excellent matches that were never visited.

Read what that equation says. Recall does not degrade uniformly; it degrades **as a function of how small the customer is**. It collapses first and hardest for exactly the tenants least able to diagnose it. And it is silent: no error, no timeout, no log line. It surfaces as “*the search feels a bit weak for us*” — a correctness bug wearing the costume of a relevance complaint, which is the disguise most likely to survive a year of triage unexamined.

The fixes are known: iterative index scans that resume the traversal until the limit is satisfied after filtering, or partitioning the index per tenant so the filter is satisfied by construction. We prefer partitioning, because it removes the failure mode rather than mitigating it — the filter becomes a question of *which index is searched* instead of a predicate applied to whatever came back.

The reason this is written down before the first embedding is computed is that it is not discoverable later. A bug that presents as a matter of taste does not get found by monitoring. It gets found by having thought about the mechanism.

5.3 *The corpus is adversarial*

A document can contain text. Text can contain instructions. A model reading a document on a tenant's behalf will read those instructions too.

The consequence is a rule with no exceptions: **the model never names the tenant**. Tools take no organisation parameter — not an optional one, not a defaulted one. The executor closes over the identity resolved from the session before any model sees anything.

If the parameter exists, prompt injection inside a document becomes a cross-tenant read: the model is persuaded to pass a different identifier and the data layer complies, because the request is perfectly well-formed. If the parameter does not exist, the same injected instruction has nothing to act upon. The capability is removed rather than guarded, and removed capabilities do not regress.

For the same reason the agent does not write queries. It gets a fixed vocabulary backed by the same functions the interface uses — so every query it can cause to run is one that already existed and was already reviewed, and the intelligence layer introduces no new path to audit.

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6 PRINCIPLES

Rules we hold to, stated so they can be held against us.

1. **The boundary is built before the intelligence.** An agent added over an unproven store inherits every gap in it and multiplies them, because a model handed a row will repeat it to whoever is listening.
2. **Make the invariant a compile error.** If correctness depends on a human remembering, it is not a property. It is a hope with good intentions.
3. **A claim without a measurement is not a claim.** When a refactor says the output is unchanged, we render both and diff the trees. When a build check says content cannot leak, we leak one on purpose and confirm the build fails. A check never observed to fail has not been tested.
4. **Fail closed, and fail visibly.** A misconfiguration that presents to the customer as an empty workspace is worse than one that raises. Both are safe; only one is honest.
5. **Store the components, not the rounded result.** Five of twelve is not fifty percent. Systems that normalise away small discrepancies lose the ability to represent them, and the discrepancy was usually the point.
6. **Determinism is a feature.** Anything generated is seeded, and the seed is pinned. An artefact that cannot be reproduced cannot be reviewed.
7. **Write the trap down where it is set.** A hazard documented at its definition is a hazard; the same hazard documented elsewhere is an outage with a delay.

8. **State the gap.** Work that was verified structurally but not on a device is described that way. An unverified claim presented as a verified one is worse than the gap it conceals.

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7 WHAT WE REFUSE

A position is defined as much by what it declines.

- **We do not train on customer data.** The asset belongs to the organisation that produced it. A company brain that quietly becomes everyone else's is not the product we described.
- **We do not provision accounts from an identity callback.** Convenient, and it means anyone who can assert an address at any provider can mint an account inside someone's organisation. Identities are matched on the provider's immutable subject, never on an email a tenant administrator is free to set.
- **We do not give the agent a query language.** Expressiveness here is indistinguishable from an unaudited data path.
- **We do not ship the workspace with the login form.** Content reaches the browser after authentication or not at all, and the build fails if that ever stops being true.

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8 THE ASSET

Every task closed, decision recorded and document written compounds into one connected account of how a company actually operates. Not a folder — a structure, in which the relationships carry as much information as the contents.

What that yields, in order of how long it takes to appear:

- People stop asking each other for status, because the state is legible without a human intermediary.
- Risk surfaces before it is a crisis, because a system watching the whole graph notices a shape that no individual, seeing one slice, can.
- What people know stops leaving when they do.
- And every model the organisation ever points at its own operation arrives with Q already high — which, by Section 3, is the only term it cannot buy.

That last point is the durable one. Model capability will keep rising, on someone else's balance sheet, available to every competitor on the same terms. The context those models are given will not. It has to be built, by the organisation, out of its own work, over time.

People come and go. The brain stays.

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